

KANAKU Mobile Platform

Native gap closure, defect remediation & release readiness

Prepared	Scope	Branch	Status
31 July 2026	Android · iOS · Auth · Ledger V2 · AA	main	Ready pending credentials

The Android app was shipping with **none of its native plugin layer compiled in**. Ten Capacitor plugins were imported by the application and zero reached the binary, so every native call — PIN storage, haptics, status bar, local notifications — rejected on device. It built, installed and launched cleanly the entire time, which is why it went unnoticed through v8. That defect is fixed, an iOS platform now exists, twenty defects are closed, and the remaining work is **entirely external**: credentials, an Apple account, and one legal declaration.

13 PLUGINS LINKED Android. Was 0 before this work.	20 DEFECTS CLOSED 6 critical, 8 high, 6 medium.	169_{/169} TESTS PASSING 22 suites. No regressions.	5 ITEMS BLOCKED All external. None on code.
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01 The headline defect

The Capacitor CLI builds its native plugin list from the `package.json` that sits beside `capacitor.config.json` — the repository root. Every plugin was declared only in `frontend/package.json`. That type-checks, bundles and imports perfectly, so nothing in CI or code review could catch it, but `cap sync` never saw the plugins and no native code was linked.

Compounding failure: the hardware back button never registered

Native setup ran under a single `try/catch`. The first call was `StatusBar`, which rejected because the plugin was absent — aborting the rest of the block, including the Android back-button handler. Every symptom downstream of that line was invisible in testing and guaranteed on device.

Fixing plugin resolution then exposed a second, latent problem: the project ran **AGP 9.2.1 / Gradle 9.6.1**, which Capacitor 8 plugin modules cannot configure under. The toolchain had been upgraded safely only because no plugin subproject was ever in the build. It is now pinned to the combination Capacitor ships and tests against.

02 Defects closed

SEVERITY	DEFECT	CONSEQUENCE ON DEVICE
Critical	Zero plugins compiled into native builds	All native calls rejected; back button never registered
Critical	AGP 9 incompatible with Capacitor 8	Every plugin module failed to configure once linked
Critical	<code>READ_SMS</code> absent from merged manifest	Permission auto-denied; SMS auto-detection silently dead
Critical	Login blocked on a 12s network budget	Full-screen spinner on every login; 42s worst case

SEVERITY	DEFECT	CONSEQUENCE ON DEVICE
● Critical	PIN keypad blocked on up to six requests	Unbounded wait before a digit could be typed
● Critical	No iOS platform existed	Config referenced an iOS target that was never generated
● High	<code>RECORD_AUDIO</code> never declared	Microphone auto-denied; voice capture failed
● High	File export used a blob URL + <code><a download></code>	Exports produced no file and no error in either WebView
● High	Notification taps had no destination	SMS alerts dead-ended; app merely raised
● High	Voice transcript processed twice per utterance	Duplicate AI calls against a quota-limited tier
● High	Export wrote to permission-gated public storage	Write rejected; export silently produced nothing
● High	CI uploaded a non-existent artifact path	Release AABs never actually published by CI
● High	Socket retried permanent auth failures	Five doomed reconnects competing with startup
● High	<code>Package.swift</code> written with Windows separators	iOS project unresolvable on macOS
● Medium	Status bar call aborted iOS native setup	Android-only API throwing on iOS
● Medium	Stray keystroke could double-submit the PIN	Second verification over a pending unlock
● Medium	"Re-sync now" synced nothing	Refreshed permissions only; never re-fetched data
● Medium	Export skipped the share sheet on native	File written where the user cannot reach it
● Medium	Splash never hidden; keyboard overlapped nav	Brand screen lingered; bottom bar covered inputs
● Medium	Dead <code>patch-package</code> scaffolding	Implied patches were applied; nothing ever ran

Two entries — permission-gated storage and the skipped share sheet — were introduced during this engagement and caught in a self-review pass before release, not found in production.

03 Capabilities delivered

CAPABILITY	ANDROID	IOS	NOTES
Biometric unlock	● Live	● Live	Face ID / Touch ID / fingerprint. Does not bypass the PIN — see §4.
Voice speech-to-text	● Live	● Pending	Plugin ships no SPM manifest; iOS falls back to the keyboard.
Push notifications	● Wired	● Wired	End-to-end; inert until credentials are supplied.
File export / share	● Live	● Live	Native share sheet replaces the no-op browser download.
Deep links	● Live	● Live	<code>kanaku://</code> registered on both; notification taps route.
App icon & splash	● Live	● Live	Generated from vector — <code>npm run assets:icons</code> .
SMS auto-detection	● Live	● N/A	Sideload flavor only. iOS exposes no SMS API — permanent.

04 Security posture

Biometrics never bypass the PIN

A biometric match unlocks a hardware-stored copy of the PIN, which then runs through the *existing* verification path — server check and key derivation unchanged. Biometrics replace typing, not verification.

The credential is stored under `BIOMETRY_CURRENT_SET`, not the plugin's default of `NONE` — which stores secrets readable with no authentication at all. It is read through a prompt cryptographically bound to the decryption key, so no code path can retrieve the PIN without a live match, and enrolling a new fingerprint destroys it.

Play Store separation holds

SMS permissions live in a flavor-scoped manifest that the Play build never merges. Both merged manifests were inspected: the sideload build carries `READ_SMS` and `RECEIVE_SMS`; the Play build carries neither, and no receiver.

Ledger V2 is inert by construction

The new publisher is gated behind `LEDGER_V2_ENABLED`, which is unset in every tracked configuration. Nothing changes in production until it is deliberately switched on.

05 Verification evidence

● Pass	TypeScript — frontend & backend	0 errors
● Pass	Unit tests — Vitest	169 / 169 · 22 suites
● Pass	Production web build — Vite	exit 0
● Pass	Android release bundles — both flavors, cold cache	618 tasks · 11.76 MB
● Pass	Merged manifests — permission separation per flavor	verified
● Pass	Deep-link parser — six input shapes	6 / 6
● Partial	Loan integration tests — database unreachable	49 / 49 *
● Not run	iOS compile — no macOS host available	CI gate

* The loan suites passed, but the staging database was unreachable from the build host, so they exercised authentication, validation and error paths rather than committing real payments. The Ledger V2 transaction refactor is type-verified and reviewed, **not** integration-verified. Re-run against a live database before enabling the feature flag.

06 Blocked — and what each needs

Nothing below is waiting on engineering.

ITEM	OWNER	REQUIRED TO UNBLOCK
Apple signing	Account admin	Developer team ID, distribution certificate, provisioning profile. CI then extends to archive & export.
Push delivery	Account admin	<code>google-services.json</code> , an APNs key, and three <code>FIREBASE_*</code> server variables.
Export compliance	Legal	A declaration, not a code change. Omitted deliberately rather than guessed.
Ledger V2 sign-off	Engineering	A reachable database to re-run the loan suites. Three further modules need schema changes — see below.
Account Aggregator	Compliance	Setu FIU onboarding, encryption key, and regulatory sign-off. Module is complete and gated.

Ledger V2 is not a wiring job — a correction to the existing plan

The roadmap described the remaining work as “add publishers”. It is not. Three of the four modules have **no cash account to post against**: investments have no account column at all, loan creation accepts no account, and goals have no server-side contribution endpoint — funding happens client-side. Only loan payments and settlements could be wired. Closing the rest requires schema migrations and API changes, each a product decision.

07 Recommended sequence

#	ACTION	WHY IT COMES HERE
1	Ship a fresh Play build	The previously published artifact predates the plugin fix and is functionally broken on device.
2	Supply Firebase & APNs credentials	Push is the largest completed feature currently delivering nothing.
3	Add Apple signing secrets	Last gate on TestFlight; icon, splash and entitlements are already in place.
4	Re-run loan suites against a live database	Only outstanding verification gap on the money path.
5	Decide the iOS voice trade-off	CocoaPods project unlocks speech-to-text; SPM is the strategic path. A judgement call, not a fix.
6	Scope Ledger V2 account modelling	Prerequisite for the remaining three modules; needs product input before engineering.

08 Reference

DOCUMENT	CONTENTS
<code>docs/release/MOBILE_RELEASE_GUIDE.md</code>	Native architecture, pinned toolchain, flavor split, build & release commands
<code>docs/release/LEDGER_V2_AND_AA_STATUS.md</code>	Publisher status, the schema blockers, pre-enablement checklist

DOCUMENT	CONTENTS
<code>docs/release/KNOWN_LIMITATIONS.md</code>	Honest register of what is partial, and why
<code>scripts/gen-app-icons.mjs</code>	Vector-sourced icon & splash generation